

VR-Forms. Preprint.

Part III. The Principle of Forms and Consistency

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2026

III.0. Introduction

Part II provided the apparatus: the formal language, register-separator, definition of a formal term, principle of forms. The present part proves a key property of this apparatus — its **safety**.

Safety means: adding the formal register to VR-Sets does not generate new contradictions and does not allow anything new to be proved in the ontological register that could not be proved in VR-Sets without the formal register. In other words, VR-Forms is a **conservative extension** of VR-Sets.

This secures the freedom of work in the formal register. One may introduce paradoxical terms, terms for uncountable collections, terms for mythological beings — and be confident that the ontological register suffers no harm. The boundary between the registers holds formally.

Contents of Part III: formal statement of conservativity (§III.1); main theorem (§III.2); consequences (§III.3); comparison with the consistency of VR-Sets relative to ZF (§III.4); summary (§III.5).

III.1. Statement

Before proving, let us precisely formulate what is being proved.

Languages and theories

Let us denote:

— L_0 — the language of VR-Sets: alphabet and terms without the marker $\ulcorner \cdot \urcorner$.

— L_1 — the language of VR-Forms: alphabet of L_0 plus the marker $\ulcorner \cdot \urcorner$, and with it the possibility of forming a formal term $\ulcorner \tau \urcorner$ from any syntactically correct τ .

— T_0 — the theory of VR-Sets: the closure principle and operational definitions (set, $\emptyset$, $\equiv$, ZFC/ZFA modes).

— T_1 — the theory of VR-Forms: everything that is in T_0 , plus the principle of forms (§II.4) and the rules of work with formal terms (§II.5).

Formulas of L_0 are called **ontological**: they speak only about operational sets. Formulas of L_1 may contain formal terms.

Definition III.1 (conservativity)

A theory T_1 is **conservative** over T_0 if for every ontological formula $\varphi \in L_0$:

$$T_1 \vdash \varphi \iff T_0 \vdash \varphi.$$

That is: everything VR-Forms proves about operational sets, VR-Sets could already have proved without the help of the formal register. The formal register adds no new ontological force.

What is to be proved

The main theorem of Part III: T_1 is conservative over T_0 .

In particular, it follows that if T_0 is consistent, so is T_1 .

III.2. Conservativity theorem

Theorem III.1 (conservativity of VR-Forms over VR-Sets)

For every ontological formula φ of the language L_0 :

$$T_1 \vdash \varphi \iff T_0 \vdash \varphi.$$

Proof

Direction ($\Leftarrow$) is trivial: every proof in T_0 is also a proof in T_1 (T_1 contains all axioms and rules of T_0).

Substantive direction ($\Rightarrow$). Let $T_1 \vdash \varphi$, where $\varphi \in L_0$. We show that $T_0 \vdash \varphi$.

Idea of the proof: every occurrence of a formal term $\ulcorner \tau \urcorner$ in the proof can be **eliminated** by replacement with an equivalent ontological formula. More precisely, formal terms are syntactic objects; their use in the proof reduces to manipulations with formulas without ontological load. If the final formula φ is ontological, all intermediate formal terms can be «crossed out» by appropriate substitution.

Step 1: translation. Define a mapping π from formulas of L_1 to formulas of L_0 :

— For a formula $\psi \in L_0$: $\pi(\psi) = \psi$.

— For an atomic formula $\ulcorner x \in \tau \urcorner$, where τ is the description $\{y : \psi(y)\}$: $\pi(\ulcorner x \in \tau \urcorner) = \psi(x)$ (comprehension, see §II.5).

— For an atomic formula $\ulcorner \tau_1 \urcorner = \ulcorner \tau_2 \urcorner$, where $\tau_1 = \{y : \psi_1(y)\}$, $\tau_2 = \{y : \psi_2(y)\}$: $\pi(\ulcorner \tau_1 \urcorner = \ulcorner \tau_2 \urcorner) = \forall y (\psi_1(y) \leftrightarrow \psi_2(y))$.

— Connectives and quantifiers — π by induction: $\pi(\neg\alpha) = \neg\pi(\alpha)$, $\pi(\alpha \wedge \beta) = \pi(\alpha) \wedge \pi(\beta)$, $\pi(\forall x \alpha) = \forall x \pi(\alpha)$, and similarly.

The translation π reduces every formula of L_1 to a formula of L_0 : formal terms are replaced by their defining predicates, and the whole formula becomes a statement about properties of objects.

Step 2: preservation of ontological formulas. If $\varphi \in L_0$, then $\pi(\varphi) = \varphi$. That is, on ontological formulas π is the identity.

Step 3: compatibility of π with the inference rules of T_1 . All inference rules of T_1 — rules of classical logic plus axioms of T_0 plus the principle of forms. We show that π translates every inference rule of T_1 into an admissible step in T_0 .

(a) **Rules of classical logic** (modus ponens, generalisation, and so on) are compatible with π , since π preserves connectives and quantifiers by definition.

(b) **Axioms of T_0** remain themselves under π — they are already in L_0 .

(c) **The principle of forms.** In T_1 the principle of forms states: every description specifies a formal term. Substantively this means that for any description $\{y : \psi(y)\}$ we can form the formal term $\ulcorner \{y : \psi(y)\} \urcorner$ and work with it by the rules of §II.5. Under π this step is translated into work directly with the predicate ψ — that is, into a usual application of the comprehension scheme in L_0 , which in T_0 generates nothing new (since the comprehension scheme in T_0 is restricted by the closure principle, see below).

Here is the subtlety. In T_0 comprehension applies to operational descriptions; in T_1 comprehension applies to arbitrary (formal). But π translates the formal term $\ulcorner \{y : \psi(y)\} \urcorner$ back into the predicate ψ . All statements about the formal term become statements about the predicate. And statements about predicates in T_0 are standard first-order statements and do not require operational describability of the predicate.

That is: the formal register serves to say «there is something described by ψ », but π converts this «something» back into simply «property ψ ». No new object appears; only a property is there, and one speaks about it in the language of T_0 .

Step 4: consequence. If $T_1 \vdash \varphi$, there is a derivation D in T_1 ending in φ . Apply π to each formula of D . We obtain a sequence of formulas of L_0 (by Step 1) in which:

— the first formula $\pi(\text{axiom of } T_1)$ is either an axiom of T_0 (if the original axiom was in T_0) or an admissible application of comprehension (if the original axiom was an instance of the principle of forms).

— every subsequent step $\pi(\text{application of rule})$ is an admissible step in T_0 (by Step 3).

— the last formula $\pi(\varphi) = \varphi$ (by Step 2).

We obtain a derivation of φ in T_0 . Hence $T_0 \vdash \varphi$.

■

Remark on the method

The method used is a standard technique for conservativity proofs via **interpretation**. The formal register receives an interpretation in the ontological: every formal term is a «figure of speech» about a property. Under such an interpretation, everything provable with the use of figures of speech is also provable without them — if the final statement is not itself a figure of speech.

This method is analogous to the interpretation of classes in NBG via formulas of ZFC, or the interpretation of abstractions in λ -calculus via β -reduction. The content is the same: a syntactic extension that does not carry new commitments.

III.3. Consequences

Corollary III.1 (consistency)

If T_0 is consistent, then T_1 is consistent.

Proof. Let T_1 be inconsistent: $T_1 \vdash \perp$. Since $\perp \in L_0$, by Theorem III.1 $T_0 \vdash \perp$. Contradiction. ■

Since the consistency of T_0 is established in VR-Sets (through the arithmetical equivalence with PA and the consistency of VR-Sets relative to ZF), it follows:

Corollary III.2

VR-Forms is consistent relative to ZF.

Corollary III.3 (safety of paradoxical formal terms)

The introduction into the formal register of the Russell term $\ulcorner \{x : x \notin x\} \urcorner$, the liar, the set of all sets, and other paradoxical descriptions does not generate contradictions in T_1 .

Justification. All these terms are syntactically correct descriptions. By the principle of forms they are formal terms. By Theorem III.1 any ontological statement derivable in T_1 with their use is derivable also without them. In particular, the contradiction $\perp \in L_0$ is not derivable in T_1 if not derivable in T_0 .

Paradoxicality here becomes a **property of the description**, not a source of contradiction. The Russell term $\ulcorner R \urcorner$ is a formal term satisfying the formula $\ulcorner R \urcorner \in \ulcorner R \urcorner \leftrightarrow \ulcorner R \urcorner \notin \ulcorner R \urcorner$. This is a formal statement about a formal term; it shows that $\ulcorner R \urcorner$ **has no ontological correlate** (if it had, the contradiction would also be in T_0). And nothing more follows from this.

Corollary III.4 (safety of non-mathematical formal terms)

The introduction into the formal register of $\ulcorner \text{dragon} \urcorner$, $\ulcorner \text{god} \urcorner$, $\ulcorner \text{soul} \urcorner$ and similar descriptions does not generate contradictions in T_1 .

Justification. Similar to Corollary III.3. These terms are syntactically correct descriptions; legitimate in the formal register by the principle of forms; their operational correlate is absent; no ontological consequences follow from their presence in the formal register.

III.4. Comparison with the consistency of VR-Sets

The chain of consistency in the VR cycle now looks like this:

VR $\leftrightarrow$ **PA** $\hookrightarrow$ **ZF** (established in VR. A Formal System, Part II)

VR-Sets $\hookrightarrow$ **ZF** (established in VR-Sets, through modelling of the operational universe in classical ZF)

VR-Forms $\hookrightarrow$ **VR-Sets** $\hookrightarrow$ **ZF** (established in the present Part III, as conservative extension)

Each link of the chain: consistency of the preceding follows from consistency of the next. If ZF is consistent, all four systems of the VR cycle are consistent.

Remark on strength. Conservativity of VR-Forms over VR-Sets — this is **equal strength** in the ontological register, but extended language. VR-Forms can speak of more (including the uncountable, paradoxical, mythological), but proves in the ontological register exactly as much as VR-Sets. This is the essence of conservative extension: extending the language without extending the theoretical strength for the original fragment of the language.

Analogy: NBG (Bernays–Gödel) is a conservative extension of ZFC, adding classes as objects. About sets, NBG proves the same as ZFC; but NBG has a more convenient language for speaking about classes. VR-Forms plays the same role relative to VR-Sets: extension of language for talking about the non-operational, without changing ontological strength for the operational.

III.5. Summary of Part III

Established:

- (1) Distinction of languages and theories of VR-Sets (L_0, T_0) and VR-Forms (L_1, T_1) (§III.1).
- (2) Definition of conservativity (Definition III.1, §III.1).
- (3) Conservativity theorem: T_1 is conservative over T_0 (Theorem III.1, §III.2). Proof via interpreting mapping π , translating formal terms into their defining predicates.
- (4) Consistency of VR-Forms relative to ZF (Corollary III.2, §III.3).
- (5) Safety of paradoxical and non-mathematical formal terms (Corollaries III.3, III.4, §III.3).
- (6) Place of VR-Forms in the chain of consistency of the VR cycle (§III.4).

What follows. Part IV is devoted to **transits** — rules by which work in the formal register may yield results in the ontological register. The conservativity theorem guarantees the safety of transits; Part IV provides their positive content: when and how the formal register is able to **help** operational proofs.